LEARNER GUIDE

Surface Area

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

The surface area of a solid is the total area of its whole outside surface — every face added together. In this lesson you take a solid, unfold it into its flat net, and find that the surface area is simply the area of that net. You work through a cuboid, a triangular prism, a square-based pyramid and a cylinder, and you watch the one tricky face — the curved side of a cylinder — peel off and unroll into a plain rectangle. The question you are exploring is: how much material would it take to cover a solid completely?

What you are learning to notice

- That the surface area of a solid is the total area of its net — every face measured once and added.
- How to find the surface area of a cuboid, a triangular prism and a square-based pyramid.
- That a cylinder's curved surface unrolls into a rectangle 2 π r wide and h tall, so its area is 2 π r h.
- That the total surface area of a closed cylinder is $2 \pi r^2 + 2 \pi r h$ — two circles plus that rectangle.

Moving through it, one step at a time

Choose a solid and press Unfold to net to watch it open out flat. Work out the area of one of each face, type it in, then add every face to find the total surface area. In Movement, drag the unroll slider to peel the curved side off the cylinder and see the rectangle it becomes. In Reflection, decide whether each calculation is right or has a slip. Then, as a packaging designer, work out how much card or paper a real object needs. Take your time, and use the Scratchpad to sketch nets and label their sides.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the area of a rectangle, a triangle and a circle (πr^2), the net of a solid, and π), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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